

Shriram M

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EDUCATION

Selvam College of Technology

Bachelor of Engineering in Computer Science and Engineering

Namakkal, TN

2022 – May 2026

TECHNICAL SKILLS

AI & Agentic Systems: Python, LangChain, Spring AI, RAG Architectures, Agent Orchestration (MCP, asyncio), Prompt Engineering, Llama 3

Backend & APIs: FastAPI, Java 17, Spring Boot 3.x, Node.js, REST/JSON-RPC, SSE Transport

Databases (Vector & SQL): ChromaDB, pgvector, PostgreSQL, Supabase, MySQL

Frontend & Cloud: React.js, Next.js, Tailwind CSS, Docker, Kubernetes, Vercel, Git

EXPERIENCE

Software Development Engineer Intern

Feb 2026 – Mar 2026

Servimos Technologies Pvt Ltd (WeServeTech LLC)

Chennai, TN

- **Automated weekly compliance reporting** by building an AI report generator in Java 17 and Spring Boot 3.x, cutting manual preparation time from ~4 hours to <10 minutes per cycle.
- **Implemented RAG-based context injection** via pgvector + PostgreSQL for semantic document retrieval and applied MCP (SSE transport, JSON-RPC tool calling) to automate recurring reporting pipelines.
- **Delivered a production-ready AI backend** over 3 Agile sprints with end-to-end ownership from schema design through deployment, with code and architecture reviewed and approved by senior engineers.
- **Established production-grade observability** for the reporting service using structured logging and Spring Boot Actuator metrics, enabling rapid incident debugging and SLA monitoring after go-live.

PROJECTS

Prompt Forge — AI Meta-Framework Synthesis Engine | *Python, FastAPI, Supabase, Groq*

2026 | Live

- **Designed a full-stack SaaS compiler** that transforms vague user inputs into structured, FAANG-grade XML prompts by dynamically evaluating complexity and applying 78+ advanced cognitive frameworks.
- **Integrated Groq's Llama 3 API** via a Python/FastAPI backend, using strict metaprompting and Best-of-N gating to reduce hallucinations while maintaining sub-200ms prompt compilation latency.
- **Hardened API security** by enforcing strict Pydantic payload validation to prevent LLM Denial-of-Wallet attacks, implementing a Debounce API to block disposable emails, and tightening CORS controls.
- **Implemented tier-based usage tracking** on Supabase with RLS and JWT auth to enforce Base vs. ML-Elite rate limits without cross-tenant data leakage.

Agentic Meeting Assistant — AI Workspace Secretary | *FastAPI, MV3, Groq (Llama-3.3)*

2026 | Source

- **Built a lightweight Javascript service worker** (Chrome MV3) to bypass dynamic Google Meet obfuscation using a greedy polling loop for live caption streaming.
- **Engineered deterministic output validation** using Python's Pydantic to convert unpredictable LLM hallucinations into strictly typed class objects to prevent downstream API crashes.
- **Designed a stateful Human-in-The-Loop SQLite database** to freeze low-confidence outputs in a PENDING state for manual review before triggering headless OAuth REST API executions across Google Workspace.

Axiom — Multi-Agent RAG Boardroom | *Python, FastAPI, Llama 3.1, ChromaDB, React*

2026 | Source

- **Architected a multi-agent RAG system** that simulates a corporate C-suite, utilizing `asyncio.gather` for concurrent, sub-second orchestration of 3 distinct LLM personas via the Groq API.
- **Engineered strict database-level security isolation** using ChromaDB metadata filtering, ensuring department-scoped knowledge retrieval and zero cross-agent data leakage.
- **Designed a shared-memory boardroom architecture** where agents dynamically evaluate, negotiate, and push back on cross-department proposals based on conflicting system prompts.

- PharmaGraph Knowledge Engine** — NL2Cypher Graph DB tool | *FastAPI, Neo4j, Groq (Llama-3)* 2026 | Source
- **Architected a production-grade Neo4j graph database** coupled with Llama-3 (Groq API) to execute zero-shot NL2Cypher queries, dynamically resolving complex pharmacological entities with high precision.
 - **Developed a scalable Graph RAG pipeline** to map advanced medical ontology queries directly into executable graph traversals, dramatically outperforming traditional multi-table SQL joins.
 - **Built a full-stack React/Vite dashboard** to natively visualize the executed analytical nodes in real-time, bridging plain language medical inquiries with robust data representations.
- Voice-Native Claim Adjudicator** — AI Concierge | *FastAPI, MCP, ElevenLabs, WebSockets* 2026 | Source
- **Engineered a full-duplex conversational AI agent** using ElevenLabs that securely authenticates strict patient identifiers (Name, DOB, Member ID) via natural audio interruption handling.
 - **Architected a secure Python FastMCP backend** that allows cloud-based LLMs to dynamically query robust local databases without ever exposing raw schemas to the model context window.
 - **Built a custom glassmorphic UI dashboard** that aggressively polls WebSocket data structures to instantly render authenticated claim results in real-time as the AI agent speaks.
- Autonomous Medical Claim Adjudicator** — Agentic RPA System | *UiPath, Gemini 2.5 API, JSON2026* | Source
- **Integrated Google's Gemini 2.5 Flash LLM** natively into a UiPath architecture via RESTful HTTP protocols to intelligently process unstructured EOB data, bypassing the strict limitations of traditional DOM scraping.
 - **Engineered a flawless web traversal loop** using simulated hardware events and fuzzy selectors to autonomously navigate proprietary portals, input patient keys, and securely extract dynamically rendered HTML blocks.
 - **Developed advanced string sanitization algorithms** and exact JSON configuration to parse multi-dimensional LLM responses and seamlessly inject AI-adjudicated decision statuses back into local arrays.
 - **Hardened the AI automation pipeline** against strict API rate limits and rendering logic using offset delays and `ContinueOnError` suppression to achieve a 100% completion rate without human intervention.
- Vision Cluster AI** — Autonomous Detection Pipeline | *Python, OpenCV, TensorFlow* 2026 | Source
- **Deployed an unsupervised ML pipeline** using custom TensorFlow inferencing that detects and semantically clusters unstructured objects across chaotic, high-density 4k image datasets.
 - **Engineered dynamic canvas scaling algorithms** ensuring pixel-perfect mathematical extraction for bounding boxes across various screen permutations and raw input streams.
 - **Secured the backend processing API** against malicious payload uploads by implementing deep Magic Byte MIME validation (Hex binary scanning) instead of naive file extension trusts.
- Active Recall Generator** — Dual-Step LLM Video Parser | *Python, Llama-3, API Security* 2026 | Source
- **Designed a fully stateless web application** that dynamically formats raw YouTube transcripts into active recall quizzes utilizing a dual-step LLM extraction constraint workflow, bypassing the need for database persistence.
 - **Mitigated complex LLM prompt injection attacks** natively within the parsing logic by enforcing highly rigorous XML boundary wrappers around all untrusted external user inputs.
 - **Hardened the production FastAPI endpoints** to prevent Denial-of-Wallet API vulnerabilities by engineering a global rate-limiting algorithm defaulting to 30 requests per minute.